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Resources

1. Concept board Link: Verification and Validation of VLA Systems –...
2. GitHub - PKU-Alignment/VLA-Arena: VLA-A...
   a. Mujoco based arena with varying task difficulties(l0-l3) measuring perofromance across tasks for safety(obstacles), generalization and robustness(distractors).
3. LIBERO-PRO: Towards Robust and Fair Evalu...
   a. we introduce LIBERO-PRO, a plug-and-play evaluation benchmark that enables systematic assessment under multiple dimensions of variation—object, position, instruction, task, and environment—as well as their arbitrary combinations,
4. Benchmarking Vision Language Action Mod...
   a.

Mean Squared Error (MSE) serves as our primary metric for evaluating model performance on offline robotics trajectories. In the context of offline reinforcement learning, MSE has proven to be a reliable metric for estimating optimal value functions and has demonstrated strong empirical performance. For our benchmark, MSE is particularly appropriate due to several key properties:

1. Non-Negativity: The metric remains non-negative, ensuring that errors are consistently accounted for without potential cancellation effects from opposing signs.
2. Sensitivity to Large Errors: The squared term in MSE emphasizes larger deviations, providing clear indication of significant prediction errors.
3. Bias-Variance Trade-off: MSE inherently captures both bias and variance components, offering a comprehensive measure of prediction accuracy.

For a given prediction, MSE is calculated as:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{1}$$

where y_i represents the ground truth action, $\hat{y}_i$ is the predicted action, and n is the number of observations.

For our benchmark, we employ MSE to evaluate how accurately models predict actions given the observation states, image observation, and language instruction at each timestep. Given the offline nature of the OpenX dataset and the inability to evaluate models on physical robots, comparing predicted and ground truth action tensors provides the most direct assessment of model performance.

We report several variations of MSE to provide a comprehensive performance analysis:

1. **Average MSE (AMSE):** Computed as the mean MSE across all timesteps in a dataset, AMSE enables direct comparison of model performance across different datasets and architectures.
2. **Normalized AMSE (NAMSE):** Calculated as an average of the min-max normalized MSEs over all the timesteps in the dataset, this metric normalizes prediction errors to each model's error range, facilitating more equitable cross-dataset for a single model comparison by accounting for different scales in model outputs and dataset action spaces.

3. **Completion Rate:** We assess successful completion by comparing final predicted actions with ground truth final actions for all episodes in the dataset. While this serves as an approximate measure of task completion, it provides valuable insights into models' ability to reach target states across trajectories.

5. Evaluation Benchmarks | jonyzhang2023/aw...
   a. Collection of papers on VLA
6. From Intention to Execution: Probing the Ge...

7. [2409.12894] VLATest: Testing and Evaluatin...
    a. Has ideas on what kind of test needs to be done to evaluate model performance
        i. Camera angle change
        ii. Lighting change
        iii. Distractor objects
        iv. Unseen objects
8. Evaluating Uncertainty and Quality of Visual...
    a. Have proposed metrics to measure uncertainty in system prediction
    b. Have metrics to measure quantitative performance(acceleration and velocity) as an indicator for system perofromance. Relation to self driving car: 2021-Jahangirova-ICST.pdf
9. Ask Before You Act: Token-Level Uncertainty for Intervention in Vision-Language-Action Models pdf
    a. Uses uncertainity(perplexity and entropy) as a measure for human intervention
    b. Can they be used as a measure of validation?
10. VLA-RISK: pdf
    a. Create adversarial conditions and test the vla performance and provide benchmarks for robustness and safety.
11. Real-to-sim Robot policy eval: paper.pdf
    a. Using gaussian splatting to build a realistic simulation which has less variance then the real-world.
12. LBM from TRI: A Careful Examination of Large Behavior Mo...
    a. They have a separate section on evaluation protocols in real-world and simulation

Uncertainity Quantification with COnformal Prediction:

https://www.youtube.com/watch?v=R1dnPAYGwnk

https://www.youtube.com/watch?v=R1dnPAYGwnk

Robots that ask for help: Conformal Prediction for LLM Planners

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https://www.youtube.com/watch?v=WnQopZKJsTw

Real-Time Out-of-Distribution Failure Prevention via Multi-Modal Reasoning

https://arxiv.org/pdf/2505.10547

From Demonstrations to Safe Deployment: Path-Consistent ...

This paper presents a novel approach to safe deployment of diffusion policies without compromising task success.

https://tum-lsy.github.io/pacs/

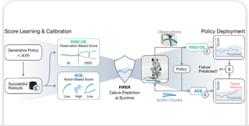
2410.04640

https://arxiv.org/pdf/2410.04640

Failure Prediction at Runtime for Generative Robot Policies (...

This paper presents a novel approach to failure prediction at runtime for generative robot policies without relying on failure data.

https://tum-lsy.github.io/fiper_website/



[2508.11117] Robot Policy Evaluation for Sim-to-Real Transfe...

Current vision-based robotics simulation benchmarks have significantly advanced robotic manipulation research. However, robotics is fundamentally a...

https://arxiv.org/abs/2508.11117

Uncertainty Comes for Free: Human-in-the-Loop Policies with Diffusion Models

https://arxiv.org/pdf/2503.01876

Fail2Progress: Learning from Real-World Robot Failures with Stein Variational Inference

https://arxiv.org/pdf/2509.01746

Uncertainty-aware Latent Safety Filters for Avoiding Out-of-Distribution Failures

https://arxiv.org/pdf/2505.00779

ARMADA: Autonomous Online Failure Detection and Human Shared Control Empower Scalable Real-world ...

https://arxiv.org/pdf/2510.02298

[2506.19121] CUPID: Curating Data your Robot Loves with In...

In robot imitation learning, policy performance is tightly coupled with the quality and composition of the demonstration data. Yet, developing a precise...

https://arxiv.org/abs/2506.19121

pdf

https://openreview.net/pdf?id=199tJ1rxmh

Policy evaluation for guided datga collection

2505.07728

https://arxiv.org/pdf/2505.07728

CORL conference papers:

RoboEval 2025 - Zoom

Zoom is the leader in modern enterprise video communications, with an easy, reliable cloud platform for video and audio conferencing, chat, and webinars...

https://cmu.zoom.us/rec/play/zQQUa30B4FQtjJP3Q9LphNUnboqaABhJv18DZGrbpgw8yC53Tp...

RobotEvaluation@RSS2025

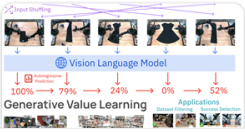
Workshop Overview Robotics has seen rapid advancements, with emerging methods and systems achieving impressive results on public benchmarks. Yet, despite these strides, real-world deployment contin...

https://sites.google.com/stanford.edu/robot-evaluation-rss-2025/

Generative Value Learning: Vision Language Models are In-...

Vision Language Models are In-Context Value Learners

https://generative-value-learning.github.io/



GitHub - villekuosmanen/physical-AI-interpretability: Attention mappers and visualisation for transformer-b...

Attention mappers and visualisation for transformer-based Physical AI policies - villekuosmanen/physical-AI-interpretability

https://github.com/villekuosmanen/physical-AI-interpretability?tab=readme-ov-file

State of Vision-Language-Action (VLA) Research at ICLR 202...

Comprehensive analysis of Vision-Language-Action models at ICLR 2026 - discrete diffusion, reasoning VLAs, and benchmark insights.

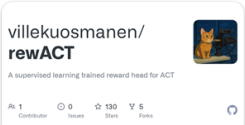
https://mbreuss.github.io/blog_post_iclr_26_vla.html

SARM: Stage-Aware Reward Modeling for Long Horizon Robot Manipulation

SARM: Stage-Aware Reward Modeling for Long Horizon Robot Manipulation

https://qianzhong-chen.github.io/sarm.github.io/

villekuosmanen/rewACT: A supervised learning trained rewa...

A supervised learning trained reward head for ACT. Contribute to villekuosmanen/rewACT development by creating an account on GitHub.

https://github.com/villekuosmanen/rewACT